PORTFOLIO PROJECT PDF

NVIDIA Risk Management Analytics ALY6130 Final Project

This is my portfolio version of my ALY6130 final project. I used public NVIDIA materials, a manually built risk register, scenario diagrams, and Monte Carlo simulation to study major business risks around supply chain concentration, export controls, and geopolitical tension.

Python | Risk Register | Heat Map | Influence Diagram | Monte Carlo

SIMPLE NOTE

This project did not use a raw transaction dataset. It was a scenario-based enterprise risk analysis case built from public company filings, risk scoring, and simulation inputs for decision support.

BUSINESS PROBLEM

NVIDIA is growing fast, but its business is exposed to supplier concentration, export controls, and geopolitical instability. My goal was to identify the most critical risks and turn them into a practical response plan that management could use.

PROJECT INFO

At a glance

Course: ALY6130 - Risk Management for Analytics

Term: Winter 2025

Type: Individual final project

Original deliverables: Final report, briefings, diagrams, notebook

Dataset: NVIDIA public filings + manual risk register + scenario inputs

Working structure: 6 final risks in one register

Main tools: Python, risk scoring sheet, diagram design

Model note: 10,000 simulations over 6 quarters

Focus risk areas: Supply chain and geopolitical tension

OVERVIEW

Project Snapshot

The original course project moved step by step from SWOT and risk identification to scoring, prioritization, scenario modeling, and response planning. This portfolio version keeps the final Module 6 story and shows the most visual parts of the work.

BUSINESS FOCUS

Enterprise risk

Map NVIDIA's main operational and external risks into a practical management view.

DATA SCALE

6 final risks

One structured risk register with scores, causes, impacts, mitigation, owner, and status.

METHODS USED

Qual + quant

Risk scoring, heat map, influence diagram, scenario tree, and Monte Carlo simulation.

DELIVERABLES

Report + visuals

Final report, briefing slides, scenario diagrams, and a Python simulation notebook.

SITUATION

NVIDIA had strong growth, but also heavy exposure to TSMC, policy risk, and Taiwan-related disruption risk.

TASK

I needed to identify the most important risks, prioritize them, and show how management choices could reduce exposure.

ACTION

I built a final risk register, converted scores into a heat map, and modeled scenario paths with Python.

RESULT

The final story became clear: supply chain disruption and geopolitical tension were the two top-priority risks.

My role in this project

- Reviewed the final report requirements and kept the project aligned with the risk management life cycle.
- Built the final risk scoring logic and organized the risk register structure.
- Used diagrams to connect decision paths, disruption paths, and sales outcomes.
- Used a simple Monte Carlo notebook to compare six-quarter sales ranges across risk paths.

Contribution note
This was an individual final project. I kept the design simple on purpose so the risk story stayed easy to explain.

STAKEHOLDERS

Executives: decide capital allocation and risk posture.

Supply chain leaders: monitor foundry concentration and delivery risk.

Trade / compliance teams: watch export policy changes.

Finance / investor relations: understand downside sales and revenue pressure.

CORE QUESTION

Which risk paths matter most, and which response path gives NVIDIA a more stable sales range over the next six quarters?

BUSINESS PROBLEM AND ANALYSIS PLAN

Problem Framing, Data Design, and Workflow

I treated this as a decision-support project. Instead of predicting one exact future number, I used structured risk logic to show which risks were more critical, how they connected, and how different management responses could change the sales range.

PROJECT PURPOSE

Turn a broad list of NVIDIA risks into one clear management story: rank the risks first, then model the highest-risk paths, then connect those paths to concrete actions.

Dataset summary	Details
Main source	NVIDIA public filings, investor materials, and course-built risk artifacts
Working structure	6-row final risk register with cause, result, mitigation, owner, and priority
Scenario input	Quarterly sales assumptions for compliance, delays, and trade restriction paths
Simulation setup	10,000 runs across 6 quarters with 0.25 / 0.50 / 0.25 scenario weights
Important note	No raw transaction table; this is a scenario-based risk analytics case

MAIN QUESTIONS

Which risks are most critical?
How should they be ranked?
What sales paths do they create?
Which response path looks more stable?

WORKFLOW

Identify risks -> score chance and impact -> build register -> visualize priority -> model response paths -> simulate six-quarter sales ranges.

TOOLS

Python for simulation, spreadsheet-style scoring logic, influence diagram and scenario tree design, and report writing.

SELECTED CODE - MONTE CARLO SETUP

```
def monte_carlo_simulation(sales_data):
    results = []
    for _ in range(10000):
        total_sales = 0
        for _ in range(6):
            scenario = np.random.choice(
                ["Calm", "Moderate", "Intense"],
                p=[0.25, 0.50, 0.25]
            )
            sales = np.random.choice(sales_data[scenario])
            total_sales += sales
        results.append(total_sales)
    return results
```

I kept the model simple. The goal was not a black-box forecast. It was to turn business assumptions into a readable downside and stability comparison.

EXAMPLE FIELDS I USED

Risk name, cause, result, consequence, chance score, impact score, risk score, priority, mitigation, owner, status.

FINAL RESPONSE PATHS MODELED

Diversify suppliers -> regulatory compliance path

Invest in other fabs -> manufacturing delays path

Invest in other fabs -> trade restrictions path

> **Why this approach fit the class**
> The course focused on enterprise risk management, not only on raw-data modeling. This workflow matched that purpose well.

FINAL SCORING STRUCTURE

Risk Scoring Sheet and Final Risk Register

I first scored each risk with a simple chance x impact approach. Then I moved the results into one final register so the project could go from ranking to action planning.

HIGHEST SCORE
63
Geopolitical tensions

SECOND HIGHEST
56
Supply chain disruptions

PRIORITY MIX
2 high
Four risks stayed in the medium tier.

REGISTER SIZE
6 risks
Each one has cause, consequence, mitigation, owner, and status.

Risk Calculation Sheet

Impact of Risk : defined as the impact to the firm before the controls. (Inherent risk)	
Very Low	1
Somewhat Low	2
Moderate	4
Somewhat High	6
High	8
Extremely High	9

List on

Operational. Includes anything to do with business operations, including supply chain, for example.	Competitive Risks: Some competitors in the industry might hurt the company's business. Pick hospitals that are closed to competitors' health center, count the total admissions change rates in these hospitals.	Reputational: May impact the company brand or standing associated with a positive or negative incident/issue	Technical. Depends on your organization.	Legal/Compliance. Risk of litigation, new regulations, or failure to comply with existing regulations, for example.

Description	Chance of Happening	Impact	Risk Score = (Chance) x (Impact)
Over-reliance on TSMC causing production delays	7	8	56
U.S. government policies reducing market sales	5	8	40
AMD and Intel improving their products	5	6	30
U.S.-China conflict disrupting supply chains	7	9	63
Competitors introducing cost-effective AI models	5	6	30
Investor concerns causing unpredictable market valuation changes	5	6	30

This scoring sheet is where the final priority logic became clear. Geopolitical tension scored 63 and supply chain disruption scored 56, so those two moved to the center of the project.

Final Risk Register

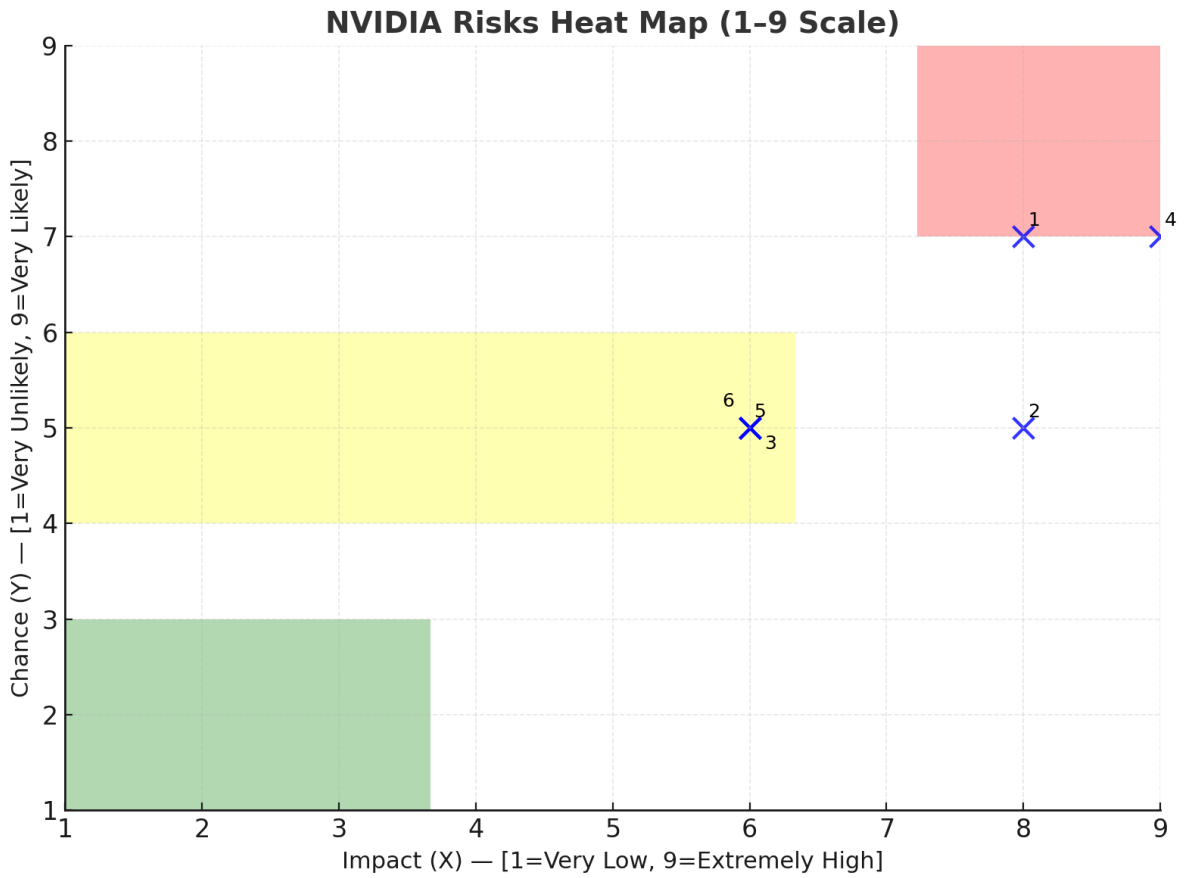
Risk #	The Risk of/That	Caused by	Resulting In	Consequence/Impact	Chance Score	Impact Score	Risk Score	Priority (H, M, L)	Mitigation	Risk Owner	Open/Closed
1	Supply chain disruptions	Over-reliance on TSMC	Production delays and lost sales	Financial and operational harm.	7	8	56	H	Partner with alternative suppliers like Samsung, Intel. Build local production facilities	Supply Chain Manager	Open
2	Export restrictions	U.S. government policies	Reduced sales in key markets	Revenue loss.	5	8	40	M	Build partnerships in safe markets and follow trade rules. Develop localized product versions	Global Trade Manager	Open
3	Strong competition	AMD and Intel improving their products	Loss of customers and price reductions	Reduced profits and weaker leadership.	5	6	30	M	Increase research and innovation to stay ahead of competitors. Accelerate GPU release cycles	R&D Manager	Open
4	Geopolitical tensions	U.S.-China conflict and Taiwan's status	Disruptions in supply chain and production	Severe global supply chain interruptions	7	9	63	H	Keep an eye on political tensions and diversify chip production to non-volatile regions	Risk Management Team	Open
5	Emergence of cost-effective AI models	Development of affordable AI by competitors	Reduced demand for NVIDIA GPUs	Loss of market share and decreased revenue.	5	6	30	M	Keep creating better and unique products and develop affordable AI solutions for mid-tier enterprises.	Strategy Manager	Open
6	Market valuation volatility	Investor concerns over competitive threats	Rapid decline in stock value	Significant financial instability.	5	6	30	M	Plan finances carefully and enhance investor transparency	Finance Manager	Open

I kept one structured table with risk name, cause, result, mitigation, owner, and open status. That made the project easier to explain as a management document, not just as an academic scoring exercise.

PRIORITY VISUAL

NVIDIA Risks Heat Map (1-9 Scale)

The heat map made the priority story easier to read. It showed that R1 and R4 were not only important, but also sat in the high-impact and high-likelihood area at the same time.



R1 = supply chain disruptions, R4 = geopolitical tensions. The other risks still mattered, but they did not sit in the same urgent zone.

WHAT THIS CHART TOLD ME

- Supply chain disruption is a direct operational risk because of heavy reliance on TSMC.
- Geopolitical tension is broader and can amplify both production and trade problems.
- Export controls matter, but they still ranked lower than the top two risks in the final scoring.
- Competition, low-cost AI models, and valuation volatility stayed in the medium-priority group.

QUICK BUSINESS READ

The project moved from a broad risk list to one clear message: focus first on supply chain resilience and geopolitical exposure.

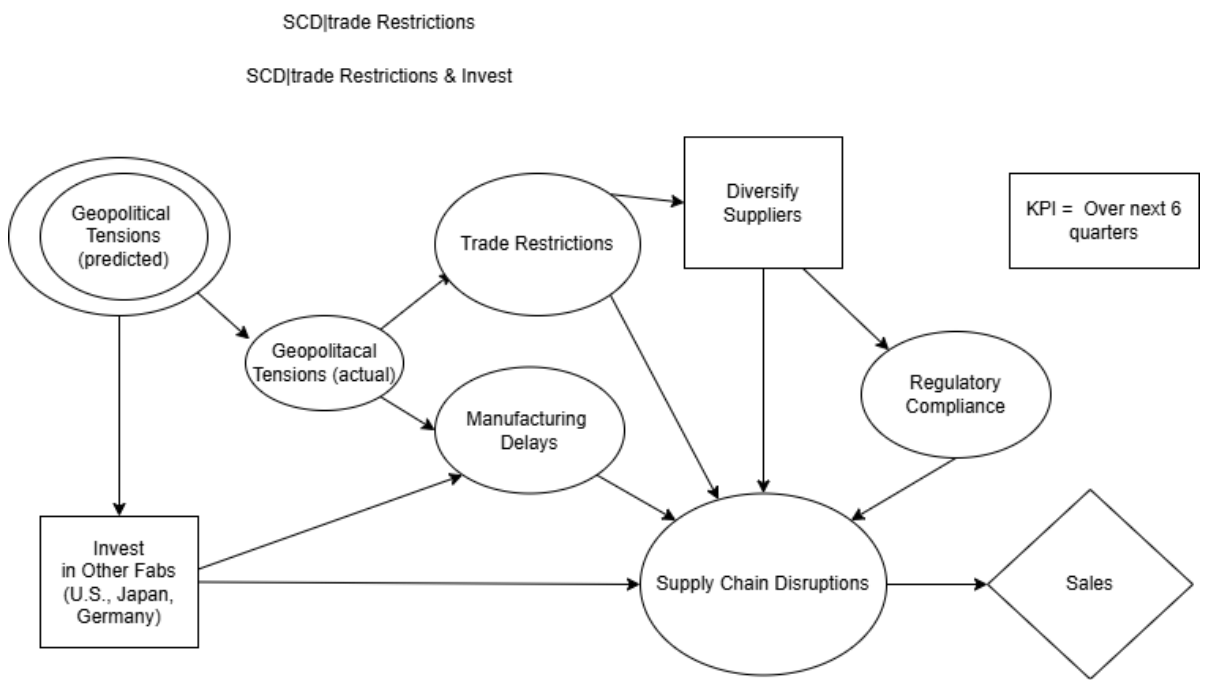
PRACTICAL IMPLICATION

If management has limited time and budget, the first response should not be spread across all six risks equally. The top two deserve earlier attention, tighter monitoring, and faster escalation rules.

DECISION SUPPORT

Influence Diagram and Response Logic

After the scoring step, I built a simple decision structure. The goal was to show how geopolitical tension could flow into manufacturing delays, trade restrictions, and supply chain disruption, and how management choices could change that path.



The diagram connects predicted geopolitical tension, actual disruption, diversify suppliers, regulatory compliance, manufacturing delays, trade restrictions, and final sales.

DECISION QUESTION

The main question was not "Can NVIDIA remove risk?" It was "Which response path creates a more stable sales range over the next six quarters?"

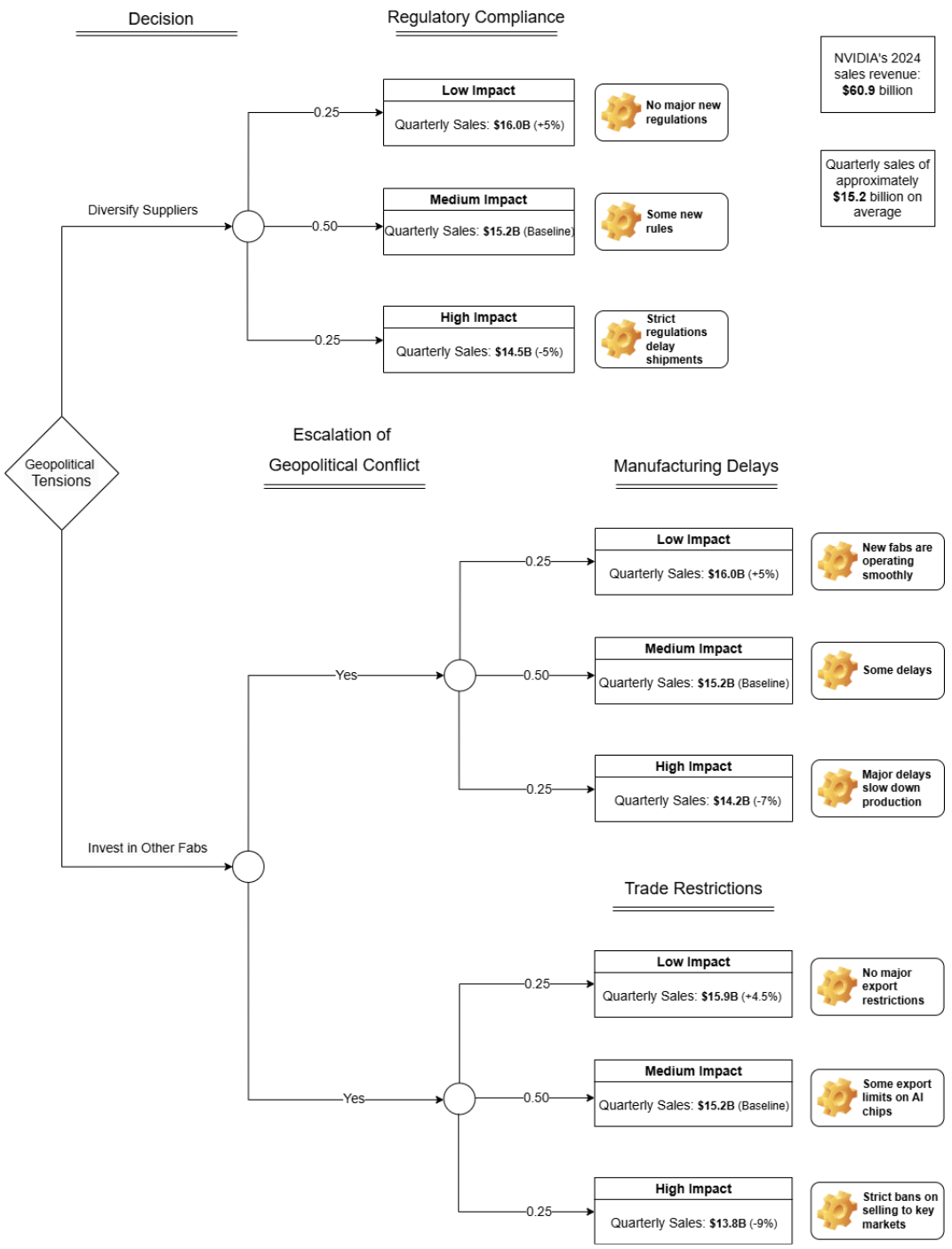
MODELED RESPONSE PATHS

Diversify suppliers for better compliance flexibility, or invest in more fabs outside Taiwan to reduce disruption exposure.

SCENARIO TREE

Updated Scenario Tree

I used NVIDIA's 2024 sales baseline and mapped three-quarter outcome levels for regulatory compliance, manufacturing delays, and trade restrictions. This made the final model easier to explain step by step.

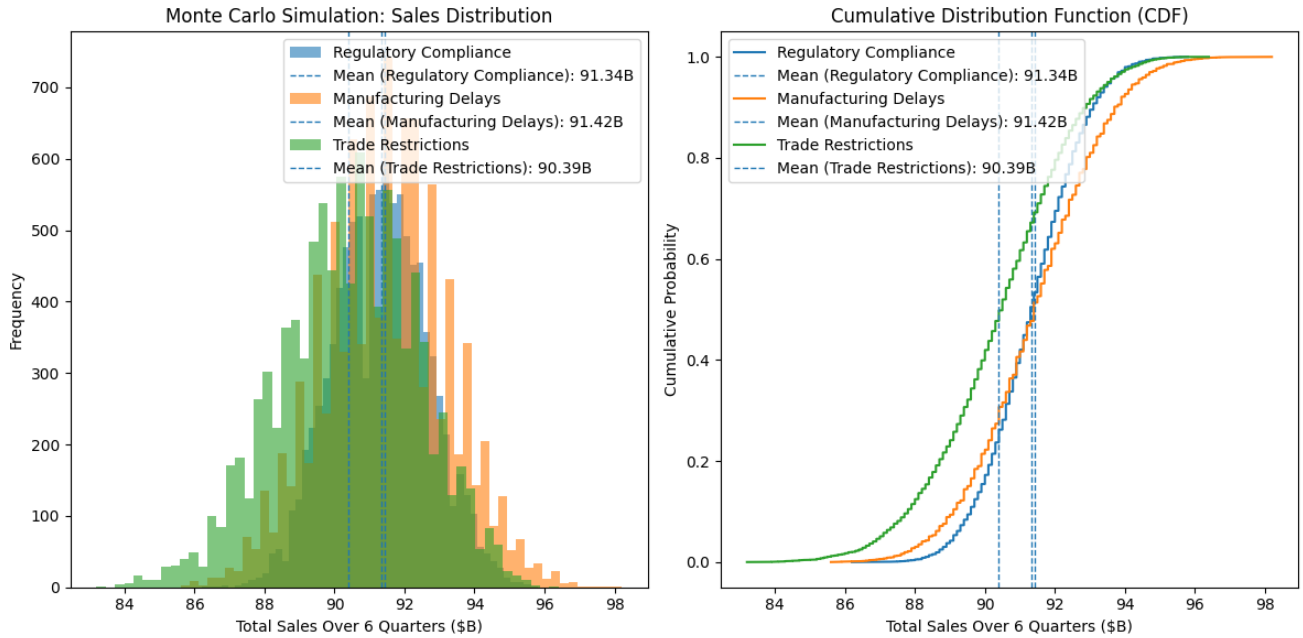


The tree starts from geopolitical tensions and branches into two management paths: diversify suppliers or invest in other fabs. Those branches then flow into regulatory compliance, manufacturing delays, and trade restrictions with low, medium, and high impact sales levels.

QUANTITATIVE RESULT

Monte Carlo Result and Executive Summary

I ran 10,000 simulation runs to compare three scenario groups: regulatory compliance, manufacturing delays, and trade restrictions. The chart helped me compare both average outcomes and downside risk.



The overlap is useful, but the left tail matters most. The trade restriction curve shifts earlier, which means worse low-end outcomes are more likely under that path.

MAIN FINDINGS

Geopolitical tensions (63) and supply chain disruption (56) were the two top risks.

Trade restrictions had the weakest downside profile in the simulation.

RESULT MEANING

Compliance looked most stable. Manufacturing delays had a wider spread. Trade restrictions carried the clearest downside tail risk.

RECOMMENDATIONS

Strengthen non-Taiwan capacity, monitor export policy shifts, and treat TSMC lead time and policy changes as early-warning KRIs.

HOW I WOULD EXPLAIN THIS IN AN INTERVIEW

Simple version: I built a risk management case around NVIDIA. I first ranked six risks, then focused on the two biggest ones, and used scenario modeling plus Monte Carlo to compare which response path gave a more stable six-quarter sales range.

WHAT THIS PROJECT SHOWS ABOUT MY WORK

- I can turn a broad business problem into a clear analytical structure.
- I can connect qualitative risk logic with quantitative comparison.
- I can organize visuals into a client-friendly story, not only a class report.
- I can keep my explanation simple and still make the business meaning clear.